

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
plt.style.use('ggplot')

#& importing file
df1= pd.read_csv('datasets.csv')

df= df1.copy()

df
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{"name":"host_id","rawType":"int64","type":"integer"},
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{"name":"neighbourhood_group","rawType":"object","type":"string"},
{"name":"neighbourhood","rawType":"object","type":"unknown"},
{"name":"latitude","rawType":"float64","type":"float"},
{"name":"longitude","rawType":"float64","type":"float"},
{"name":"room_type","rawType":"object","type":"unknown"},
{"name":"price","rawType":"float64","type":"float"},
{"name":"minimum_nights","rawType":"float64","type":"float"},
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{"name":"last_review","rawType":"object","type":"unknown"},
{"name":"reviews_per_month","rawType":"float64","type":"float"},
{"name":"calculated_host_listings_count","rawType":"float64","type":"float"},
{"name":"availability_365","rawType":"float64","type":"float"},
{"name":"number_of_reviews_ltm","rawType":"float64","type":"float"},
{"name":"license","rawType":"object","type":"string"},
{"name":"rating","rawType":"object","type":"string"},
{"name":"bedrooms","rawType":"object","type":"string"},
{"name":"beds","rawType":"int64","type":"integer"},
{"name":"baths","rawType":"object","type":"string"}],"ref":"9feee769-994c-46ed-87fe-ae8b37c7f3f1","rows":[[{"0","1312228.0","Rental unit in Brooklyn · ★5.0 · 1 bedroom","7130382","Walter","Brooklyn","Clinton Hill","40.68371","-73.96461","Private room","55.0","30.0","3.0","20/12/15","0.03","1.0","0.0","0.0","No License","5","1","1","Not specified"],["1","45277537.0","Rental unit in New York · ★4.67 · 2 bedrooms · 1 bed · 1 bath","51501835","Jeniffer","Manhattan","Hell's Kitchen","40.76661","-73.9881","Entire home/apt","144.0","30.0","9.0","01/05/23","0.24","139.0","364.0","2.0","No License","4.67","2","1","1"],["2","9.71e+17","Rental unit in New York · ★4.17 · 1 bedroom · 2 beds · 1

```

bath","528871354","Joshua","Manhattan","Chelsea","40.75076357","-73.99460515","Entire home/apt","187.0","2.0","6.0","18/12/23","1.67","1.0","343.0","6.0","Exempt","4.17","1","2","1"],["3","3857863.0","Rental unit in New York · ★4.64 · 1 bedroom · 1 private bath","19902271","John And Catherine","Manhattan","Washington Heights","40.8356","-73.9425","Private room","120.0","30.0","156.0","17/09/23","1.38","2.0","363.0","12.0","No License","4.64","1","1","1"],["4","40896611.0","Condo in New York · ★4.91 · Studio · 1 bed · 1 bath","61391963","Stay With Vibe","Manhattan","Murray Hill","40.75112","-73.9786","Entire home/apt","85.0","30.0","11.0","03/12/23","0.24","133.0","335.0","3.0","No License","4.91","Studio","1","1"],["5","49584983.0","Rental unit in New York · ★5.0 · 1 bedroom · 1 bath","51501835","Jeniffer","Manhattan","Hell's Kitchen","40.75995","-73.99296","Entire home/apt","115.0","30.0","5.0","29/07/23","0.16","139.0","276.0","2.0","No License","5","1","1","1"],["6","45277537.0","Rental unit in New York · ★4.67 · 2 bedrooms · 1 bed · 1 bath","51501835","Jeniffer","Manhattan","Hell's Kitchen","40.76661","-73.9881","Entire home/apt","144.0","30.0","9.0","01/05/23","0.24","139.0","364.0","2.0","No License","4.67","2","1","1"],["7","9.71e+17","Rental unit in New York · ★4.17 · 1 bedroom · 2 beds · 1 bath","528871354","Joshua","Manhattan","Chelsea","40.75076357","-73.99460515","Entire home/apt","187.0","2.0","6.0","18/12/23","1.67","1.0","343.0","6.0","Exempt","4.17","1","2","1"],["8","3857863.0","Rental unit in New York · ★4.64 · 1 bedroom · 1 private bath","19902271","John And Catherine","Manhattan","Washington Heights","40.8356","-73.9425","Private room","120.0","30.0","156.0","17/09/23","1.38","2.0","363.0","12.0","No License","4.64","1","1","1"],["9","40896611.0","Condo in New York · ★4.91 · Studio · 1 bed · 1 bath","61391963","Stay With Vibe","Manhattan","Murray Hill","40.75112","-73.9786","Entire home/apt","85.0","30.0","11.0","03/12/23","0.24","133.0","335.0","3.0","No License","4.91","Studio","1","1"],["10","49584983.0","Rental unit in New York · ★5.0 · 1 bedroom · 1 bath","51501835","Jeniffer","Manhattan","Hell's Kitchen","40.75995","-73.99296","Entire home/apt","115.0","30.0","5.0","29/07/23","0.16","139.0","276.0","2.0","No License","5","1","1","1"],["11","45457047.0","Rental unit in New York · ★4.33 · Studio · 1 bed · 1 bath","51501835","Jeniffer","Manhattan","Hell's Kitchen","40.76737","-73.98787","Entire home/apt","105.0","30.0","3.0","31/08/22","0.1","139.0","364.0","0.0","No License","4.33","Studio","1","1"],["12","45847112.0","Rental unit in New York · ★4.50 · 2 bedrooms · 2 beds · 1 bath","51501835","Jeniffer","Manhattan","Hell's Kitchen","40.76519","-

73.99013", "Entire
home/apt", "130.0", "30.0", "10.0", "30/05/23", "0.26", "139.0", "295.0", "2.0", "No License", "4.5", "2", "2", "1"], ["13", "19280212.0", "Rental unit in Brooklyn · ★4.79 · 1 bedroom · 1 bed · 1 shared
bath", "2526182", "Margaux", "Brooklyn", "Williamsburg", "40.71153", "-73.95312", "Private
room", "90.0", "30.0", "19.0", "01/10/23", "0.24", "2.0", "5.0", "2.0", "No License", "4.79", "1", "1", "1"], ["14", "8.96e+17", "Rental unit in Brooklyn · ★4.67 · 1 bedroom · 1 bed · 1
bath", "14251313", "Kristin", "Brooklyn", "Sunset Park", "40.65891327", "-73.98958973", "Entire
home/apt", "292.0", "30.0", "12.0", "19/10/23", "1.71", "1.0", "365.0", "12.0", "No License", "4.67", "1", "1", "1"], ["15", "17367484.0", "Rental unit in Queens · 1 bedroom · 2 beds · 1 shared
bath", "84109314", "Mary", "Queens", "Astoria", "40.76068", "-73.90926", "Private
room", "120.0", "30.0", "1.0", "21/08/23", "0.22", "1.0", "365.0", "1.0", "No License", "No rating", "1", "2", "1"], ["16", "22066430.0", "Rental unit in Brooklyn · ★4.71 · Studio · 1 bed · 1
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home/apt", "160.0", "30.0", "49.0", "08/10/23", "0.67", "1.0", "0.0", "7.0", "No License", "4.71", "Studio", "1", "1"], ["17", "9.3e+17", "Rental unit in New York · 1 bedroom · 1 bed · 1
bath", "21662972", "Youning", "Manhattan", "Financial District", "40.70999354", "-74.00499963", "Entire
home/apt", "100.0", "30.0", "1.0", "30/08/23", "0.23", "1.0", "38.0", "1.0", "No License", "No rating", "1", "1", "1"], ["18", "26448665.0", "Rental unit in Queens · 1 bedroom · 1 bed · 1
bath", "38713773", "Fabrizio", "Queens", "Sunnyside", "40.74347", "-73.91396", "Private
room", "70.0", "30.0", "1.0", "03/11/19", "0.02", "1.0", "0.0", "0.0", "No License", "No rating", "1", "1", "1"], ["19", "28224871.0", "Rental unit in New York · ★3.20 · 2 bedrooms · 3 beds · 1
bath", "187148732", "Yehudah", "Manhattan", "Washington Heights", "40.83969", "-73.93861", "Entire
home/apt", "164.0", "30.0", "5.0", "13/04/19", "0.08", "1.0", "0.0", "0.0", "No License", "3.2", "2", "3", "1"], ["20", "29923599.0", "Serviced apartment in New York City · Studio · 2 beds · 1 bath", "220229838", "Chamber Hotel", "Manhattan", "Midtown", "40.76176", "-73.97659", "Hotel
room", "1000.0", "30.0", "1.0", "02/07/19", "0.02", "10.0", "363.0", "0.0", "Empty", "No rating", "Studio", "2", "1"], ["21", "23053827.0", "Home in East Elmhurst · 1 bedroom · 1 bed · 1 shared
bath", "171107511", "Paola", "Queens", "East Elmhurst", "40.76529", "-73.86361", "Private
room", "45.0", "30.0", "1.0", "07/10/18", "0.02", "1.0", "0.0", "0.0", "No License", "No rating", "1", "1", "1"], ["22", "34718504.0", "Condo in New York · 2 bedrooms · 2 beds · 2
baths", "6674394", "Eri", "Manhattan", "Upper West Side", "40.79283", "-

73.9719", "Entire
home/apt", "425.0", "180.0", "1.0", "25/06/19", "0.02", "1.0", "358.0", "0.0",
"No License", "No rating", "2", "2", "2"], ["23", "39022484.0", "Serviced
apartment in New York · ★4.80 · 1 bedroom · 1 bed · 1 private
bath", "241889662", "Incentra Village Guest House", "Manhattan", "West
Village", "40.73888", "-74.00343", "Private
room", "196.0", "30.0", "5.0", "26/07/22", "0.1", "12.0", "0.0", "0.0", "No
License", "4.8", "1", "1", "1"], ["24", "34511726.0", "Rental unit in
Brooklyn · ★4.65 · 1 bedroom · 1 bed · 1
bath", "260475429", "Zahira", "Brooklyn", "Kensington", "40.64304", "-
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home/apt", "120.0", "30.0", "26.0", "15/03/20", "0.47", "2.0", "0.0", "0.0", "N
o License", "4.65", "1", "1", "1"], ["25", "35376453.0", "Rental unit in
Brooklyn · 1 bedroom · 2 beds · 1
bath", "234383668", "Federica", "Brooklyn", "Williamsburg", "40.71336", "-
73.95495", "Entire
home/apt", "100.0", "30.0", "1.0", "10/08/19", "0.02", "1.0", "0.0", "0.0", "No
License", "No rating", "1", "2", "1"], ["26", "34621385.0", "Rental unit in
Brooklyn · ★5.0 · 2 bedrooms · 2 beds · 1
bath", "3899139", "David", "Brooklyn", "Bushwick", "40.70477", "-
73.92671", "Entire
home/apt", "220.0", "30.0", "12.0", "05/01/20", "0.21", "1.0", "0.0", "0.0", "N
o License", "5", "2", "2", "1"], ["27", "36819630.0", "Rental unit in
Brooklyn · ★4.75 · 1 bedroom · 1 bed · 1
bath", "188207570", "Claire", "Brooklyn", "Red Hook", "40.67673", "-
74.00194", "Entire
home/apt", "84.0", "30.0", "4.0", "12/12/20", "0.08", "1.0", "0.0", "0.0", "No
License", "4.75", "1", "1", "1"], ["28", "35574095.0", "Rental unit in New
York · 1 bedroom · 1 bed · 1 private
bath", "7258900", "Jill", "Manhattan", "Tribeca", "40.71446", "-
74.00826", "Private
room", "200.0", "30.0", "1.0", "14/06/19", "0.02", "1.0", "0.0", "0.0", "No
License", "No rating", "1", "1", "1"], ["29", "9.51e+17", "Hotel in New York
· 1 bedroom · 1 bed · 1
bath", "215565541", "LuxUrban", "Manhattan", "Hell's
Kitchen", "40.7603457", "-73.9877968", "Private
room", "261.0", "1.0", "2.0", "22/11/23", "0.5", "35.0", "0.0", "2.0", "Exempt",
"No rating", "1", "1", "1"], ["30", "40658238.0", "Townhouse in Queens · 1
bedroom · 2 beds · 1 shared
bath", "150022684", "Hoang", "Queens", "Woodside", "40.74376", "-
73.90008", "Private
room", "85.0", "30.0", "1.0", "03/01/20", "0.02", "1.0", "0.0", "0.0", "No
License", "No rating", "1", "2", "1"], ["31", "39558009.0", "Rental unit in
New York · 1 bedroom · 1 bed · 1.5
baths", "303941435", "David", "Manhattan", "Roosevelt
Island", "40.76381", "-73.94959", "Private
room", "70.0", "30.0", "1.0", "20/01/20", "0.02", "1.0", "0.0", "0.0", "No
License", "No rating", "1", "1", "1.5"], ["32", "39899915.0", "Rental unit in
The Bronx · ★4.89 · 1 bedroom · 1 bed · 1 shared

bath","307330962","Juan","Bronx","Concourse Village","40.82859","-73.91957","Private room","60.0","30.0","9.0","02/01/22","0.18","1.0","2.0","0.0","No License","4.89","1","1","1"],["33","7070705.0","Rental unit in Brooklyn · ★4.43 · 1 bedroom · 1 bed · 1 shared

bath","37072479","Joseph","Brooklyn","Williamsburg","40.71905","-73.95937","Private room","90.0","30.0","7.0","02/01/19","0.09","3.0","0.0","0.0","No License","4.43","1","1","1"],["34","5.94e+17","Rental unit in New York · 1 bedroom · 1 bed · 1

bath","4068562","Dominique","Manhattan","Harlem","40.82378","-73.9508","Entire home/apt","75.0","30.0","2.0","31/05/23","0.17","1.0","0.0","2.0","No License","No rating","1","1","1"],["35","45938560.0","Rental unit in Brooklyn · ★5.0 · 1 bedroom · 1 bed · 1 shared

bath","6913051","Ivan","Brooklyn","Williamsburg","40.71265","-73.96539","Private room","106.0","30.0","8.0","15/09/23","0.22","1.0","15.0","5.0","No License","5","1","1","1"],["36","46156111.0","Rental unit in Brooklyn · 1 bedroom · 1 bed · 2 shared

baths","3223938","Eugene","Brooklyn","Flatbush","40.65442","-73.95565","Private room","36.0","90.0","1.0","01/12/20","0.03","589.0","0.0","0.0","No License","No rating","1","1","2"],["37","9.53e+17","Hotel in Brooklyn · ★4.33 · 1 bedroom · 1 bed · 1

bath","215565541","LuxUrban","Brooklyn","Williamsburg","40.707258","-73.9339721","Private room","220.0","1.0","3.0","18/12/23","0.94","35.0","0.0","3.0","Exempt","4.33","1","1","1"],["38","48194167.0","Rental unit in The Bronx · 1 bedroom · 1 bed · 1 shared

bath","388818764","Mildred","Bronx","University Heights","40.85615","-73.9152","Private room","55.0","30.0","2.0","21/03/21","0.06","1.0","0.0","0.0","No License","No rating","1","1","1"],["39","6.62e+17","Home in Queens · ★4.82 · 1 bedroom · 1 bed · 1 private

bath","45400201","Shauna","Queens","Flushing","40.75972","-73.81527","Private room","69.0","30.0","28.0","22/12/23","1.54","6.0","55.0","12.0","No License","4.82","1","1","1"],["40","235951.0","Condo in New York · ★4.69 · 1 bedroom · 1 bed · 1

bath","1236070","Daniel","Manhattan","Midtown","40.75408","-73.97071","Entire home/apt","188.0","30.0","155.0","01/12/23","1.04","1.0","91.0","6.0","No License","4.69","1","1","1"],["41","48298348.0","Rental unit in Queens · ★5.0 · 1 bedroom · 1 bed · 1 private

bath","309612914","Abeer","Queens","East Elmhurst","40.76366","-73.86133","Private room","99.0","30.0","6.0","20/02/22","0.18","6.0","0.0","0.0","No License","5","1","1","1"],["42","367810.0","Loft in Brooklyn · ★5.0 ·

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2 bedrooms · 1 bed · 1 shared
bath","1728785","Veronika","Brooklyn","Greenpoint","40.73626","-
73.95634","Private
room","180.0","30.0","11.0","19/04/23","0.12","1.0","3.0","1.0","No
License","5","2","1","1"],["43","271694.0","Rental unit in New York ·
★4.11 · 1 bedroom · 1 bed · 1
bath","1387370","James","Manhattan","Midtown","40.75282","-
73.97315","Entire
home/apt","125.0","365.0","19.0","08/09/15","0.13","1.0","0.0","0.0","
No License","4.11","1","1","1"],["44","13213869.0","Rental unit in New
York · ★4.90 · 1 bedroom · 1 bed · Shared half-
bath","38284609","Emily","Manhattan","Washington
Heights","40.84387","-73.94179","Private
room","85.0","30.0","10.0","02/01/20","0.11","1.0","0.0","0.0","No
License","4.9","1","1","1.5"],["45","13325767.0","Rental unit in
Brooklyn · ★4.33 · 1 bedroom · 1 bed · 1 shared
bath","10864311","Habib","Brooklyn","Clinton Hill","40.69257","-
73.96887","Private
room","75.0","30.0","9.0","26/08/19","0.11","1.0","0.0","0.0","No
License","4.33","1","1","1"],["46","48676880.0","Rental unit in Queens
· 1 bedroom · 1 bed · 1 private
bath","349342033","Andrea","Queens","Ridgewood","40.70349","-
73.90875","Private
room","100.0","30.0","1.0","24/04/21","0.03","1.0","0.0","0.0","No
License","No rating","1","1","1"],["47","47942202.0","Home in Brooklyn
· ★4.50 · 1 bedroom · 1 bed · 1
bath","43825074","Masud","Brooklyn","Cypress Hills","40.68693","-
73.87378","Private
room","38.0","30.0","4.0","01/09/23","0.12","7.0","285.0","2.0","No
License","4.5","1","1","1"],["48","8.81e+17","Home in Queens · ★4.67 ·
Studio · 3 beds · 1 shared
bath","500477805","Rebecca","Queens","Flushing","40.75713694","-
73.81348864","Private
room","121.0","30.0","3.0","12/06/23","0.39","3.0","364.0","3.0","No
License","4.67","Studio","3","1"],["49","39125813.0","Rental unit in
Staten Island · ★5.0 · 1 bedroom · 3 beds · 1
bath","77290831","Nas","Staten Island","Prince's Bay","40.52339","-
74.20517","Entire
home/apt","118.0","30.0","3.0","25/06/22","0.15","1.0","179.0","0.0","
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```

```
df.info()
```

```

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RangeIndex: 20770 entries, 0 to 20769
Data columns (total 22 columns):

```

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1	name	20770 non-null	object

2	host_id	20770	non-null	int64
3	host_name	20770	non-null	object
4	neighbourhood_group	20770	non-null	object
5	neighbourhood	20763	non-null	object
6	latitude	20763	non-null	float64
7	longitude	20763	non-null	float64
8	room_type	20763	non-null	object
9	price	20736	non-null	float64
10	minimum_nights	20763	non-null	float64
11	number_of_reviews	20763	non-null	float64
12	last_review	20763	non-null	object
13	reviews_per_month	20763	non-null	float64
14	calculated_host_listings_count	20763	non-null	float64
15	availability_365	20763	non-null	float64
16	number_of_reviews_ltm	20763	non-null	float64
17	license	20770	non-null	object
18	rating	20770	non-null	object
19	bedrooms	20770	non-null	object
20	beds	20770	non-null	int64
21	baths	20770	non-null	object

dtypes: float64(10), int64(2), object(10)

memory usage: 3.5+ MB

```
df.isna().any().sum()
```

```
np.int64(12)
```

```
df.sample(10)
```

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      "type": "string"
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    {
      "name": "availability_365",
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      "type": "float"
    },
    {
      "name": "number_of_reviews_ltm",
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      "type": "float"
    },
    {
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      "rawType": "object",
      "type": "string"
    },
    {
      "name": "rating",
      "rawType": "object",
      "type": "string"
    },
    {
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      "type": "string"
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  ]
}
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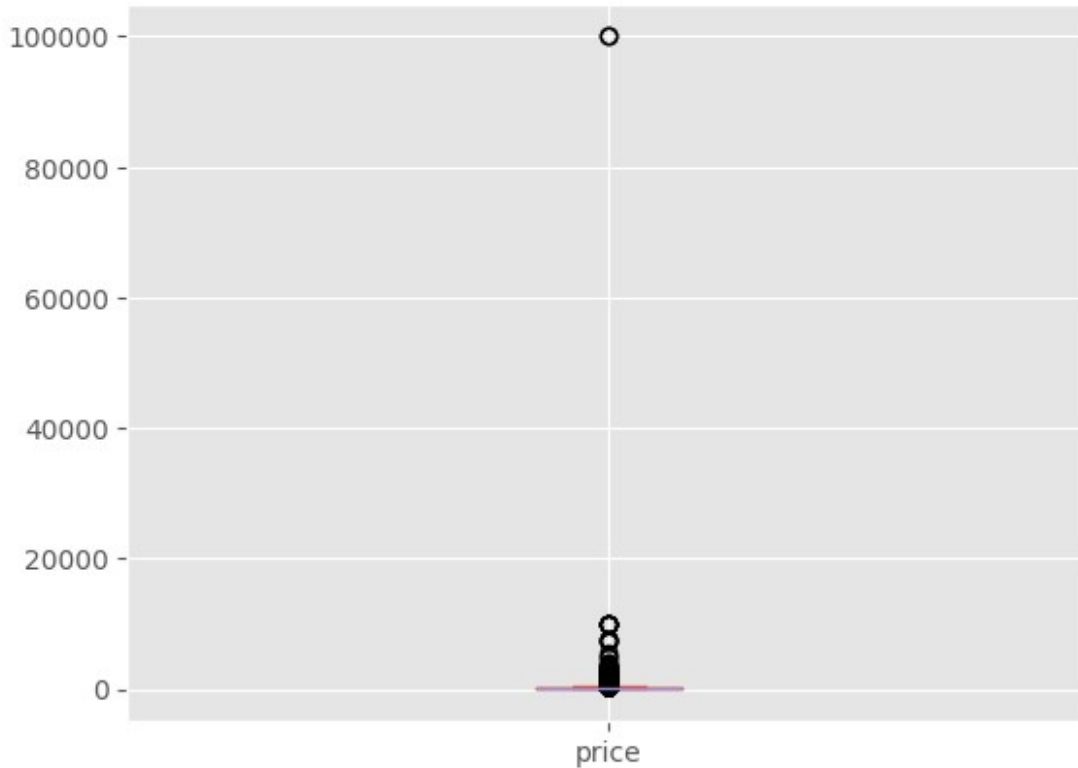
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#!/ data describe
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```
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```

```
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```

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```

```
df['price'].skew()
```

```
np.float64(89.97557856732496)
```

```
## data cleaning
```

```
df.isnull().sum()
```

```
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```

```
df[df['latitude'].isnull()]
```

```
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```

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```

drop the null values

```
df.dropna(subset=['latitude', 'last_review', 'minimum_nights', 'longitude', 'price'], inplace=True)
```

```
df.isnull().sum()
```

```
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    },
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  "shape": {
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```

```
df.shape
```

```
(20736, 22)
```

```
#! dealing with duplicated records
```

```
df[df.duplicated()]
```

```
#! delete the duplicate value
```

```
df.drop_duplicates(inplace=True)
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
#! change price to float to int
```

```
df['price'] = df['price'].astype('int')
```

```
#! changing the id and host id type of now
```

```
df['host_id'] = df['host_id'].astype('int')
```

```
df['id'] = df['id'].astype('int')
```

```
#! date time
```

```
df['last_review'] = pd.to_datetime(df['last_review'])
```

```
df
```

```
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    }
  ]
}
```

```

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Elmhurst","40.76529","-73.86361","Private
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· 2 beds · 2 baths","6674394","Eri","Manhattan","Upper West
Side","40.79283","-73.9719","Entire
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· ★4.80 · 1 bedroom · 1 bed · 1 private bath","241889662","Incentra
Village Guest House","Manhattan","West Village","40.73888","-
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· 1 bath","260475429","Zahira","Brooklyn","Kensington","40.64304","-
73.97964","Entire home/apt","120","30.0","26.0","2020-03-15
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["25","35376453","Rental unit in Brooklyn · 1 bedroom · 2 beds · 1
bath","234383668","Federica","Brooklyn","Williamsburg","40.71336","-
73.95495","Entire home/apt","100","30.0","1.0","2019-10-08
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rating","1","2","1"],["26","34621385","Rental unit in Brooklyn · ★5.0
· 2 bedrooms · 2 beds · 1
bath","3899139","David","Brooklyn","Bushwick","40.70477","-
73.92671","Entire home/apt","220","30.0","12.0","2020-05-01
00:00:00","0.21","1.0","0.0","0.0","No License","5","2","2","1"],
["27","36819630","Rental unit in Brooklyn · ★4.75 · 1 bedroom · 1 bed
· 1 bath","188207570","Claire","Brooklyn","Red Hook","40.67673","-
74.00194","Entire home/apt","84","30.0","4.0","2020-12-12
00:00:00","0.08","1.0","0.0","0.0","No License","4.75","1","1","1"],
["28","35574095","Rental unit in New York · 1 bedroom · 1 bed · 1
private bath","7258900","Jill","Manhattan","Tribeca","40.71446","-
74.00826","Private room","200","30.0","1.0","2019-06-14
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bedroom · 1 bed · 1 bath","215565541","LuxUrban","Manhattan","Hell's
Kitchen","40.7603457","-73.9877968","Private
room","261","1.0","2.0","2023-11-22
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bath","150022684","Hoang","Queens","Woodside","40.74376","-
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bedroom · 1 bed · 1.5

baths","303941435","David","Manhattan","Roosevelt
Island","40.76381","-73.94959","Private room","70","30.0","1.0","2020-
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bath","307330962","Juan","Bronx","Concourse Village","40.82859","-
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["33","7070705","Rental unit in Brooklyn · ★4.43 · 1 bedroom · 1 bed ·
1 shared
bath","37072479","Joseph","Brooklyn","Williamsburg","40.71905","-
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· 1 bedroom · 1 bed · 1 shared
bath","6913051","Ivan","Brooklyn","Williamsburg","40.71265","-
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["36","46156111","Rental unit in Brooklyn · 1 bedroom · 1 bed · 2
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["38","48194167","Rental unit in The Bronx · 1 bedroom · 1 bed · 1
shared bath","388818764","Mildred","Bronx","University
Heights","40.85615","-73.9152","Private room","55","30.0","2.0","2021-
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73.81527","Private room","69","30.0","28.0","2023-12-22
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["40","235951","Condo in New York · ★4.69 · 1 bedroom · 1 bed · 1
bath","1236070","Daniel","Manhattan","Midtown","40.75408","-
73.97071","Entire home/apt","188","30.0","155.0","2023-01-12
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Elmhurst","40.76366","-73.86133","Private
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["42", "367810", "Loft in Brooklyn · ★5.0 · 2 bedrooms · 1 bed · 1 shared bath", "1728785", "Veronika", "Brooklyn", "Greenpoint", "40.73626", "-73.95634", "Private room", "180", "30.0", "11.0", "2023-04-19 00:00:00", "0.12", "1.0", "3.0", "1.0", "No License", "5", "2", "1", "1"],
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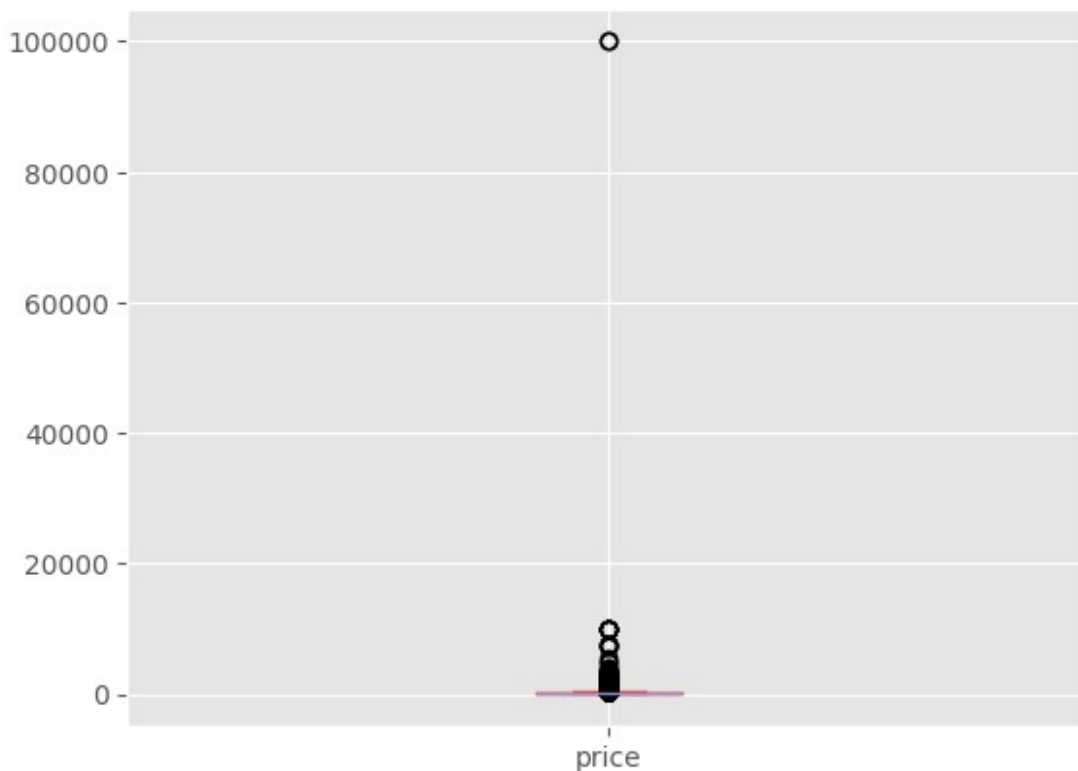
```
df.shape
```

```
(20724, 22)
```

Univariate analysis

```
df['price'].plot(kind='box')
```

```
<Axes: >
```



```
df['price'].skew()
```

```
np.float64(89.9500076528325)
```

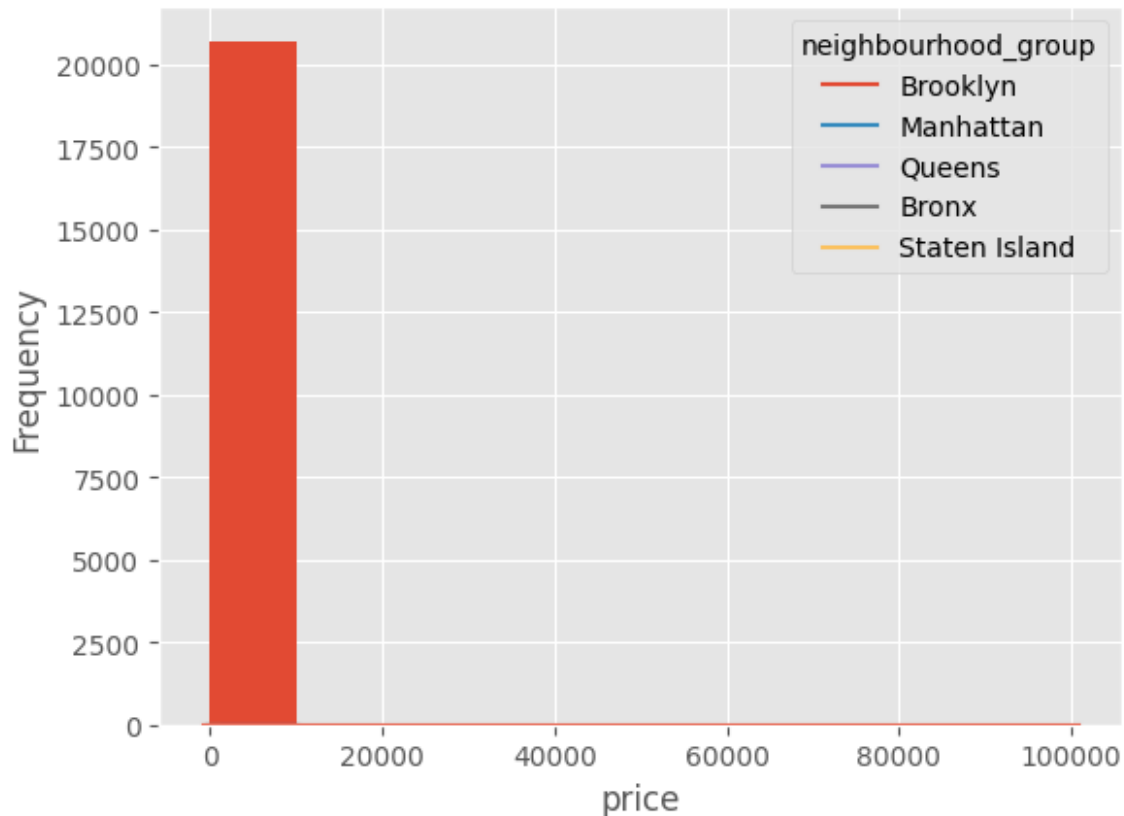
```
df[df['price']>50000]
```

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  ],
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}
```

```
df['price'].plot(kind='hist')
```

```
sns.kdeplot(data=df, x='price', hue='neighbourhood_group')
```

```
<Axes: xlabel='price', ylabel='Frequency'>
```



```
df[df['price']>10000]
```

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```



```
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      "Bobbi",
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      "1",
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}
```

```
## price analysis
# d =df[df['price']>15000].dropna(inplace=True)
```

```
## identifying outliers in price
df=df[df['price']<1200]
```

```
df.shape
```

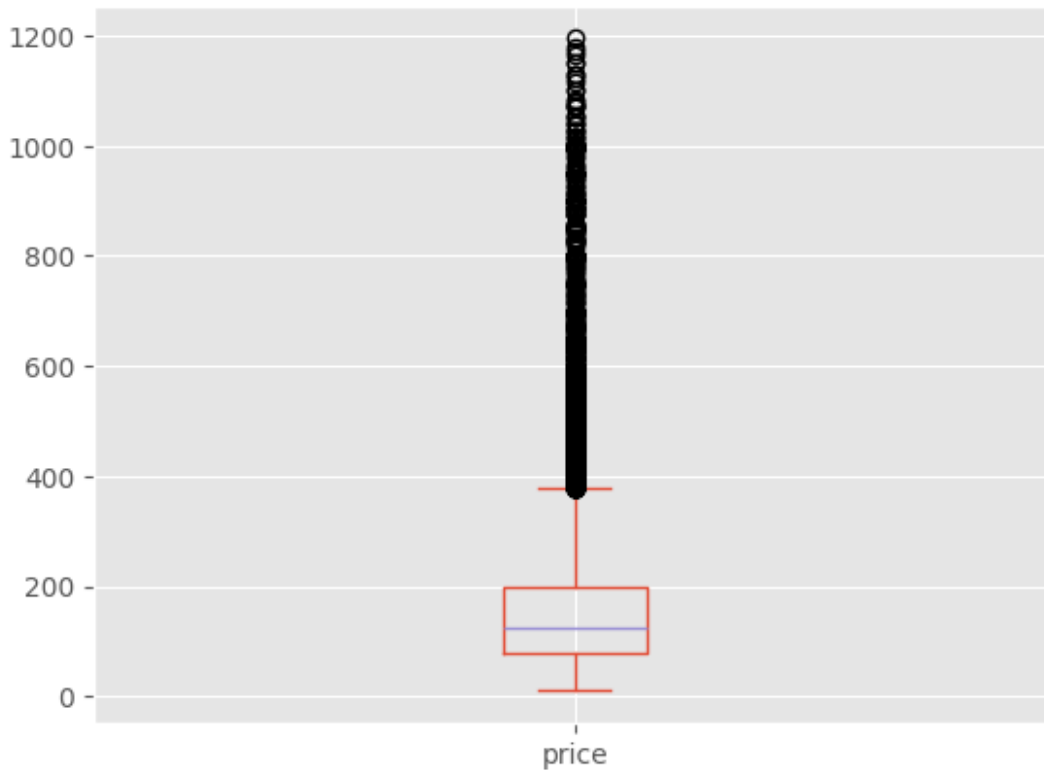
```
(20590, 22)
```

```
df['price'].sample(10)
```

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```

```
df['price'].plot(kind='box')
```

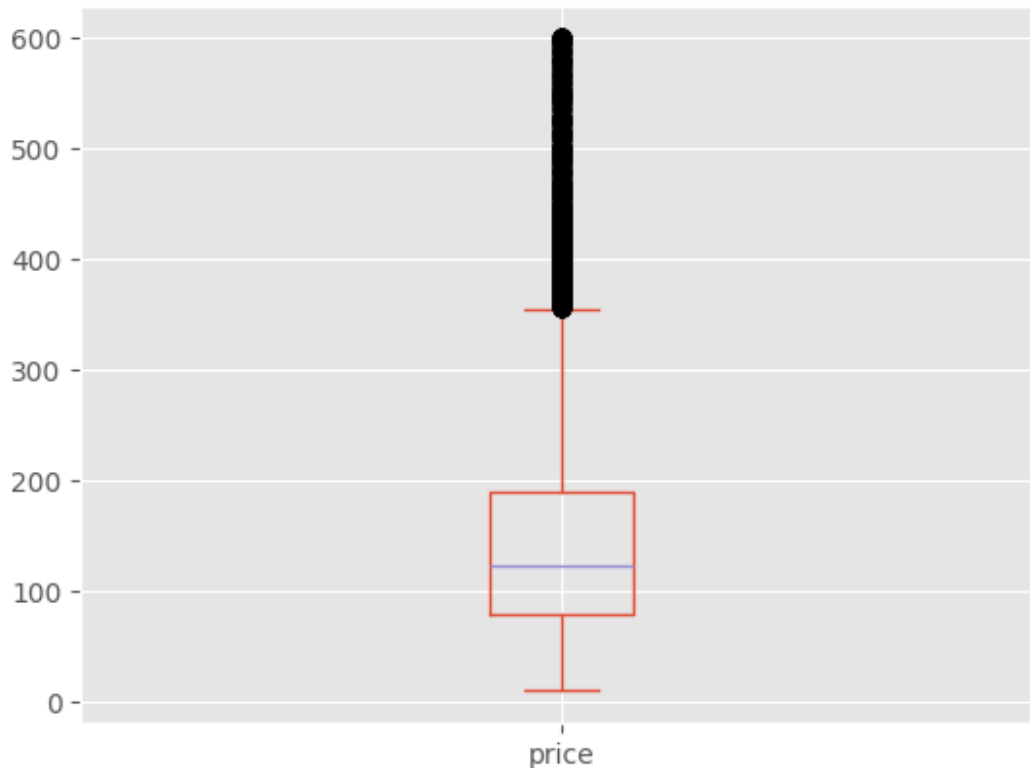
```
<Axes: >
```



```
df['price'].skew()  
np.float64(2.9071885092018275)
```

Removing outlier using upper limit one new thing

```
#!/ removing outliers  
upper_limit = df['price'].quantile(0.98)  
df= df[df['price'] <=upper_limit]  
df['price'].skew()  
np.float64(1.6385074790875476)  
df['price'].plot(kind='box')  
<Axes: >
```



df

```
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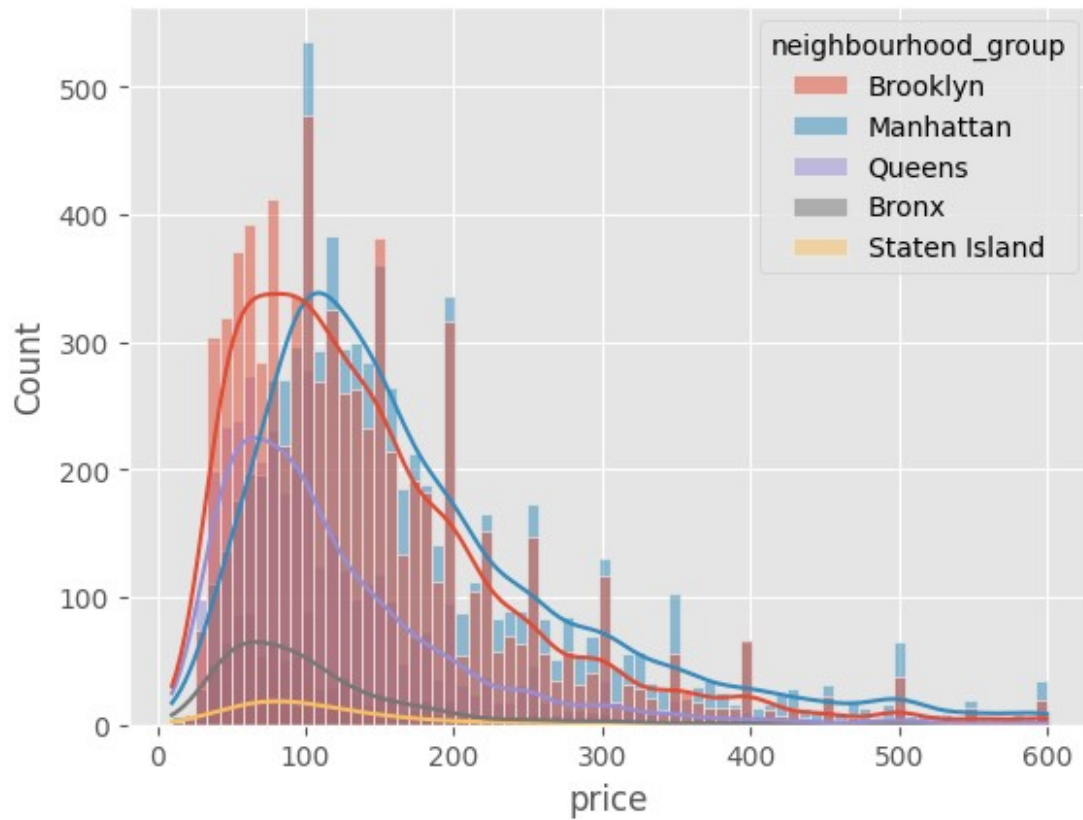
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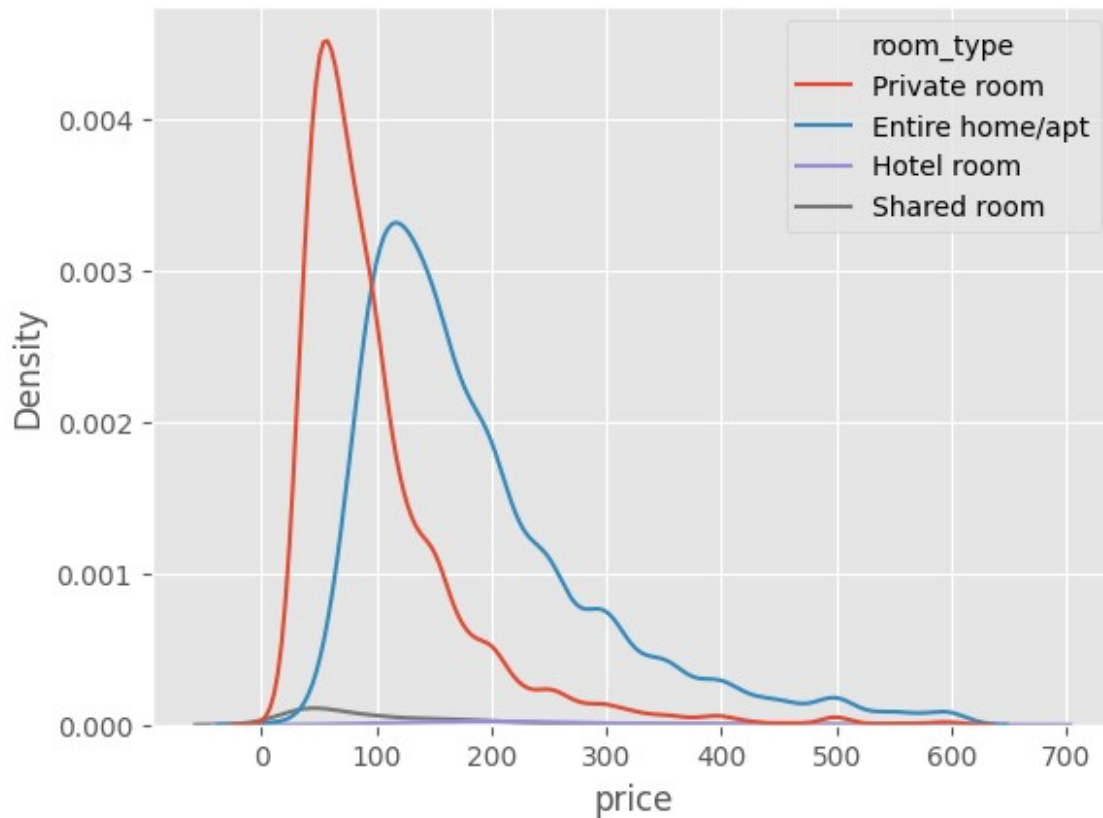
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rating","1","2","1"],["26","34621385","Rental unit in Brooklyn · ★5.0
· 2 bedrooms · 2 beds · 1
bath","3899139","David","Brooklyn","Bushwick","40.70477","-
73.92671","Entire home/apt","220","30.0","12.0","2020-05-01
00:00:00","0.21","1.0","0.0","0.0","No License","5","2","2","1"],
["27","36819630","Rental unit in Brooklyn · ★4.75 · 1 bedroom · 1 bed
· 1 bath","188207570","Claire","Brooklyn","Red Hook","40.67673","-
74.00194","Entire home/apt","84","30.0","4.0","2020-12-12
00:00:00","0.08","1.0","0.0","0.0","No License","4.75","1","1","1"],
["28","35574095","Rental unit in New York · 1 bedroom · 1 bed · 1
private bath","7258900","Jill","Manhattan","Tribeca","40.71446","-
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bedroom · 1 bed · 1 bath","215565541","LuxUrban","Manhattan","Hell's
Kitchen","40.7603457","-73.9877968","Private
room","261","1.0","2.0","2023-11-22
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["30","40658238","Townhouse in Queens · 1 bedroom · 2 beds · 1 shared
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73.90008","Private room","85","30.0","1.0","2020-03-01
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rating","1","2","1"],["31","39558009","Rental unit in New York · 1
bedroom · 1 bed · 1.5
baths","303941435","David","Manhattan","Roosevelt
Island","40.76381","-73.94959","Private room","70","30.0","1.0","2020-
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rating","1","1","1.5"],["32","39899915","Rental unit in The Bronx ·
★4.89 · 1 bedroom · 1 bed · 1 shared
bath","307330962","Juan","Bronx","Concourse Village","40.82859","-
73.91957","Private room","60","30.0","9.0","2022-02-01
00:00:00","0.18","1.0","2.0","0.0","No License","4.89","1","1","1"],
["33","7070705","Rental unit in Brooklyn · ★4.43 · 1 bedroom · 1 bed ·
1 shared
bath","37072479","Joseph","Brooklyn","Williamsburg","40.71905","-
73.95937","Private room","90","30.0","7.0","2019-02-01
00:00:00","0.09","3.0","0.0","0.0","No License","4.43","1","1","1"],
["34","594000000000000000","Rental unit in New York · 1 bedroom · 1
bed · 1 bath","4068562","Dominique","Manhattan","Harlem","40.82378","-
73.9508","Entire home/apt","75","30.0","2.0","2023-05-31
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· 1 bedroom · 1 bed · 1 shared
bath","6913051","Ivan","Brooklyn","Williamsburg","40.71265","-
73.96539","Private room","106","30.0","8.0","2023-09-15
00:00:00","0.22","1.0","15.0","5.0","No License","5","1","1","1"],
["36","46156111","Rental unit in Brooklyn · 1 bedroom · 1 bed · 2
shared baths","3223938","Eugene","Brooklyn","Flatbush","40.65442","-

73.95565","Private room","36","90.0","1.0","2020-01-12
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★4.33 · 1 bedroom · 1 bed · 1
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73.9339721","Private room","220","1.0","3.0","2023-12-18
00:00:00","0.94","35.0","0.0","3.0","Exempt","4.33","1","1","1"],
["38","48194167","Rental unit in The Bronx · 1 bedroom · 1 bed · 1
shared bath","388818764","Mildred","Bronx","University
Heights","40.85615","-73.9152","Private room","55","30.0","2.0","2021-
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★4.82 · 1 bedroom · 1 bed · 1 private
bath","45400201","Shauna","Queens","Flushing","40.75972","-
73.81527","Private room","69","30.0","28.0","2023-12-22
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73.97071","Entire home/apt","188","30.0","155.0","2023-01-12
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Elmhurst","40.76366","-73.86133","Private
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bath","1728785","Veronika","Brooklyn","Greenpoint","40.73626","-
73.95634","Private room","180","30.0","11.0","2023-04-19
00:00:00","0.12","1.0","3.0","1.0","No License","5","2","1","1"],
["43","271694","Rental unit in New York · ★4.11 · 1 bedroom · 1 bed ·
1 bath","1387370","James","Manhattan","Midtown","40.75282","-
73.97315","Entire home/apt","125","365.0","19.0","2015-08-09
00:00:00","0.13","1.0","0.0","0.0","No License","4.11","1","1","1"],
["44","13213869","Rental unit in New York · ★4.90 · 1 bedroom · 1 bed
· Shared half-bath","38284609","Emily","Manhattan","Washington
Heights","40.84387","-73.94179","Private
room","85","30.0","10.0","2020-02-01
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["45","13325767","Rental unit in Brooklyn · ★4.33 · 1 bedroom · 1 bed
· 1 shared bath","10864311","Habib","Brooklyn","Clinton
Hill","40.69257","-73.96887","Private room","75","30.0","9.0","2019-
08-26 00:00:00","0.11","1.0","0.0","0.0","No
License","4.33","1","1","1"],["46","48676880","Rental unit in Queens ·
1 bedroom · 1 bed · 1 private
bath","349342033","Andrea","Queens","Ridgewood","40.70349","-
73.90875","Private room","100","30.0","1.0","2021-04-24
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rating","1","1","1"],["47","47942202","Home in Brooklyn · ★4.50 · 1

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bedroom · 1 bed · 1 bath", "43825074", "Masud", "Brooklyn", "Cypress
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· ★4.67 · Studio · 3 beds · 1 shared
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1 bath", "39553567", "Bianca", "Staten Island", "Port
Richmond", "40.62866", "-74.14578", "Entire
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· 2 bedrooms · 1 bed · 2 baths", "5655889", "East Village
Loft", "Manhattan", "East Village", "40.72154", "-73.98174", "Entire
home/apt", "450", "30.0", "54.0", "2023-05-11
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1 bed · 1 bath", "6455628", "Teddy", "Queens", "Astoria", "40.76299", "-
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```

```

#!/ minimum night

```

```

df['availability_365'].plot(kind='hist', edgecolor='black', color='skybl
ue')

```

```

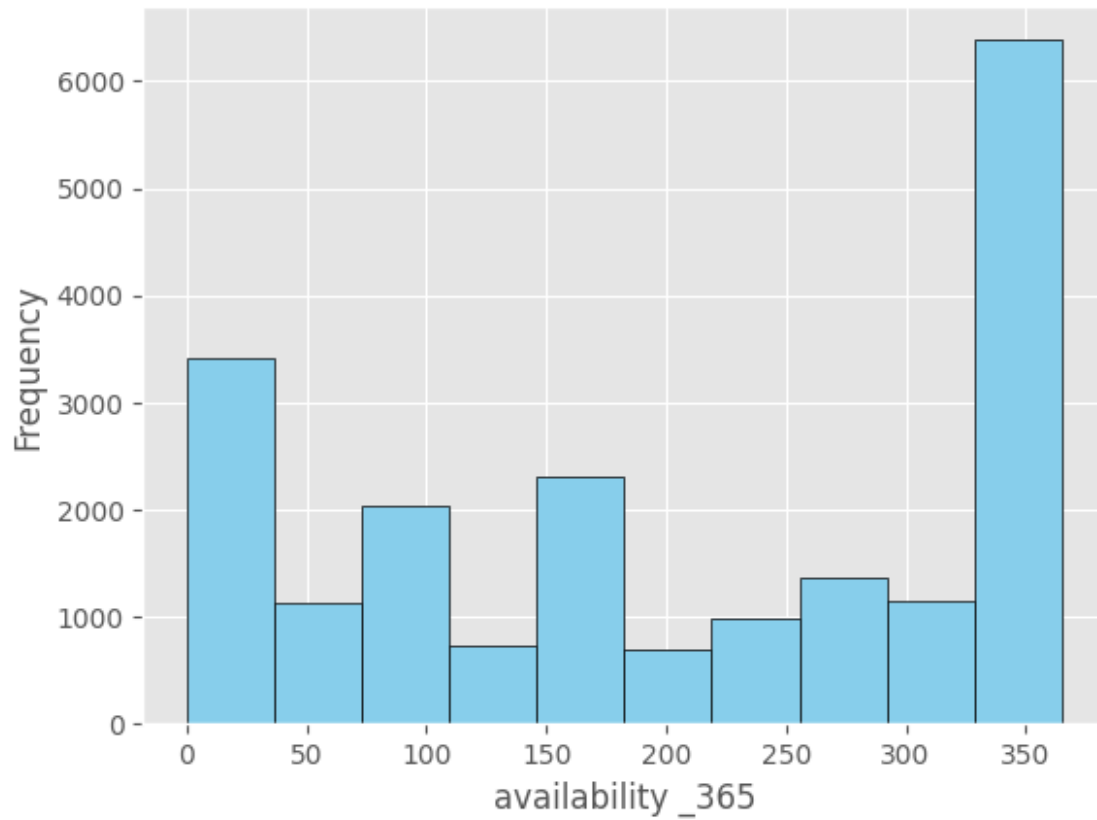
plt.xlabel('availability_365')

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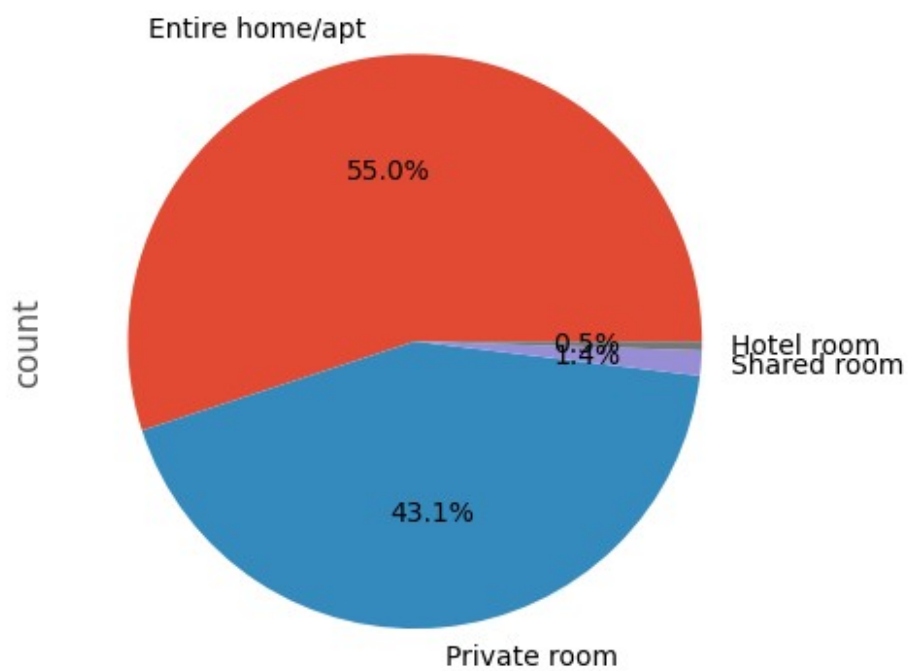
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Text(0.5, 0, 'availability_365')

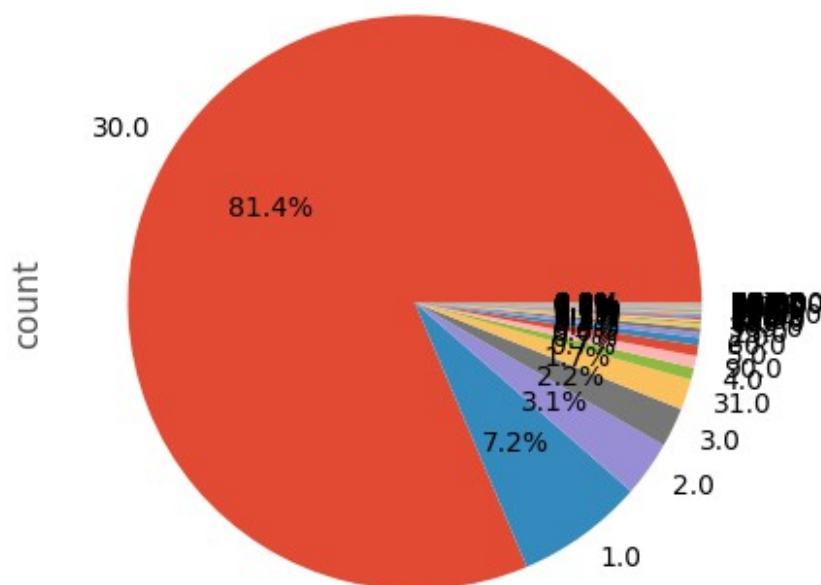
```



```
df['room_type'].value_counts().plot(kind='pie', autopct='%0.1f%%')  
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```



```
df['minimum_nights'].value_counts().plot(kind='pie', autopct="%0.1f%%")
<Axes: ylabel='count'>
```



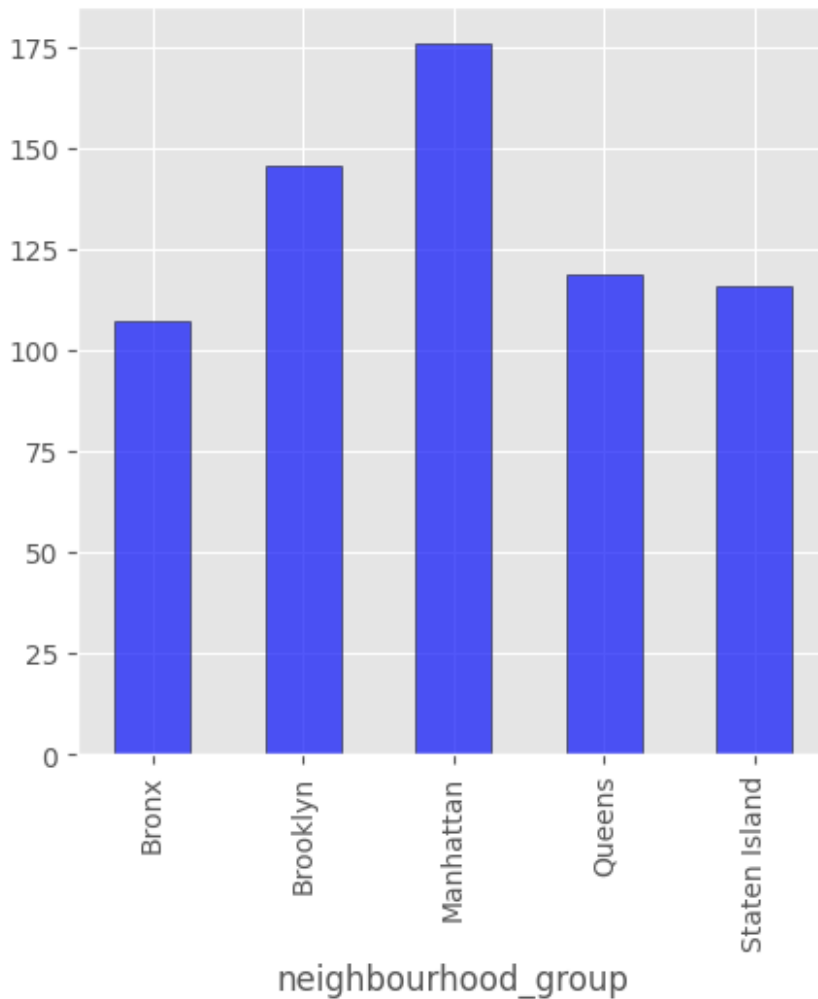
```
df.columns
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      'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',
      'minimum_nights', 'number_of_reviews', 'last_review',
      'reviews_per_month', 'calculated_host_listings_count',
      'availability_365', 'number_of_reviews_ltm', 'license',
      'rating',
      'bedrooms', 'beds', 'baths'],
      dtype='object')
```

bivariate analysis

```
#!/ neighbour hood and price group by to annalyzed

df.groupby('neighbourhood_group')
['price'].mean().plot(kind='bar',color="#0911F9",edgecolor='#070A13',f
igsize=(5,5),alpha=0.7)

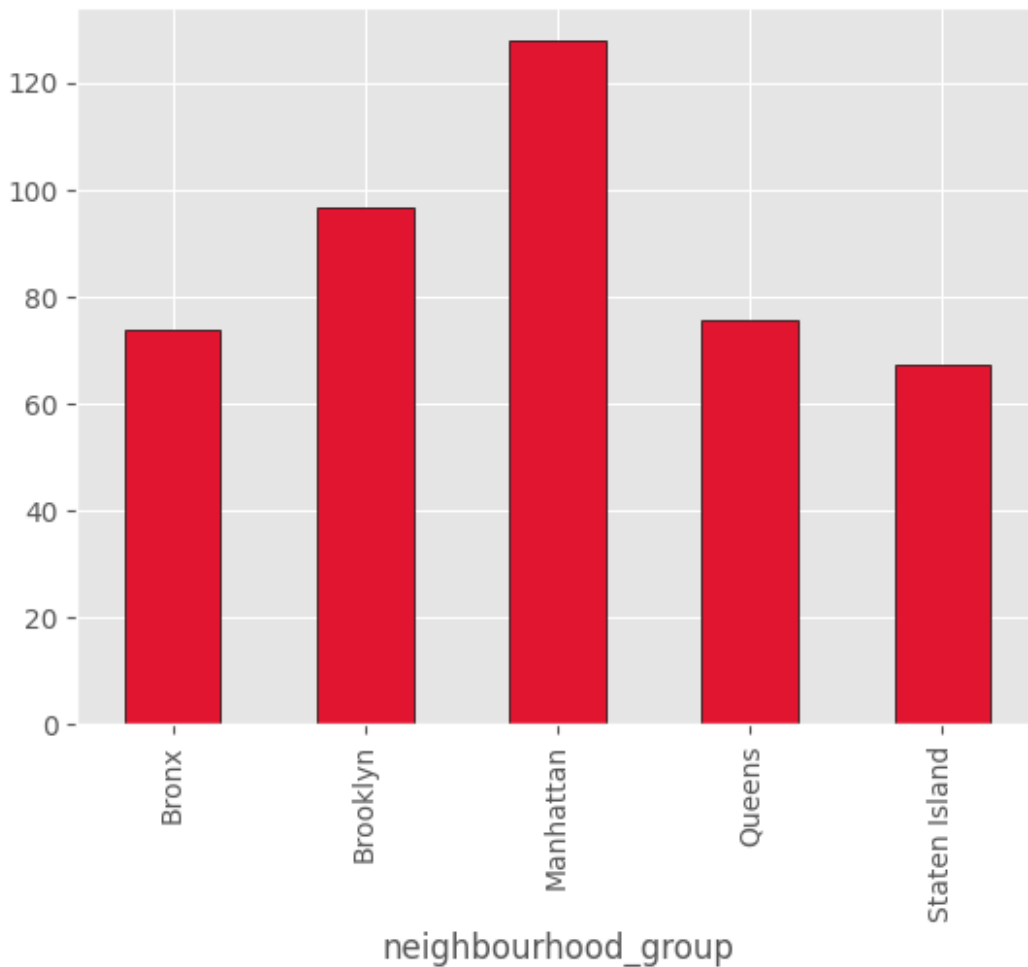
<Axes: xlabel='neighbourhood_group'>
```



per bed Feature Engineering

```
df['per_bed'] = df['price'] / df['beds']  
df.groupby('neighbourhood_group')['per_bed'].mean().reset_index()  
  
{"columns": [{"name": "index", "rawType": "int64", "type": "integer"},  
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{"1", "Brooklyn", "96.7660333710763"},  
{"2", "Manhattan", "127.73095554311426"},  
{"3", "Queens", "75.71210122082013"}, {"4", "Staten  
Island", "67.27199096880132"}], "shape": {"columns": 2, "rows": 5}}  
  
df.groupby('neighbourhood_group')  
['per_bed'].mean().plot(kind='bar', color="#E11530", edgecolor='black')
```

```
<Axes: xlabel='neighbourhood_group'>
```



bivariate analysis

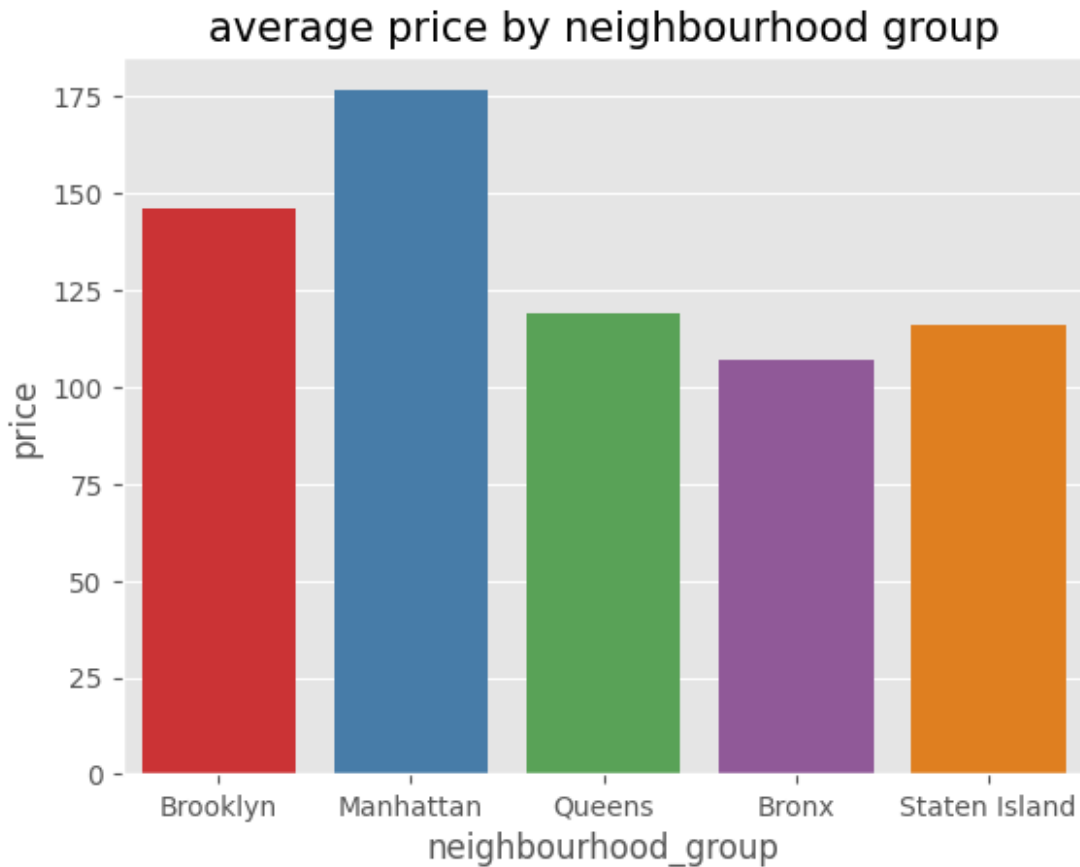
```
df.columns
```

```
Index(['id', 'name', 'host_id', 'host_name', 'neighbourhood_group',  
      'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',  
      'minimum_nights', 'number_of_reviews', 'last_review',  
      'reviews_per_month', 'calculated_host_listings_count',  
      'availability_365', 'number_of_reviews_ltm', 'license',  
      'rating',  
      'bedrooms', 'beds', 'baths', 'per_bed'],  
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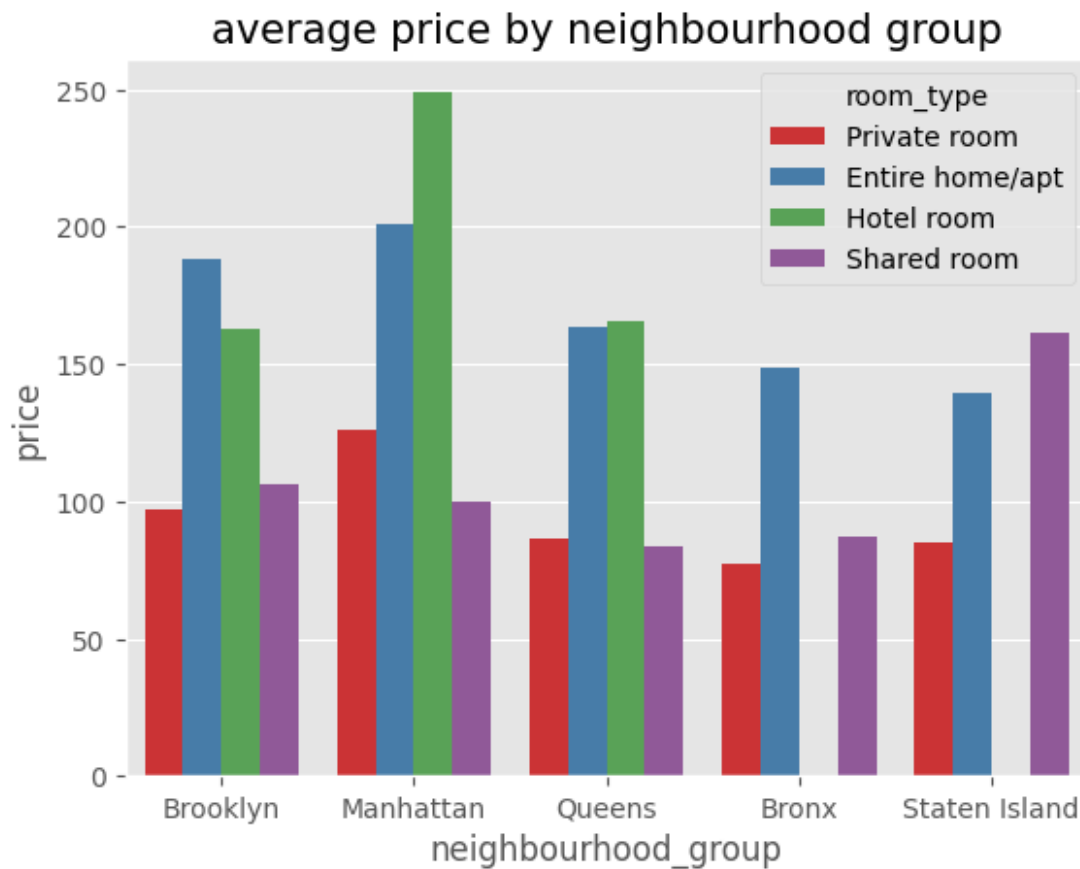
```
sns.barplot(data=df, x='neighbourhood_group', y='price', errorbar=None, es  
timator=np.mean, hue='neighbourhood_group', palette='Set1')  
plt.title("average price by neighbourhood_group")
```



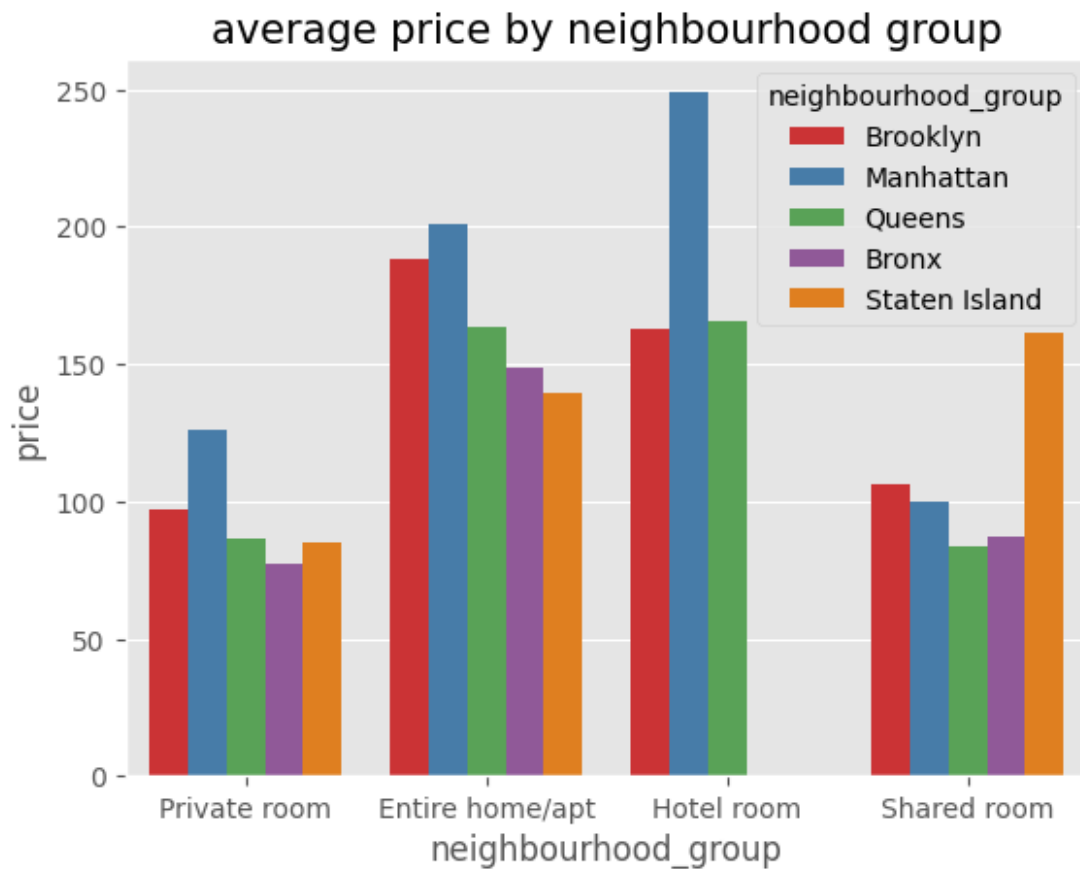
```
group", fontsize='15', weight='light')
plt.xlabel("neighbourhood_group")
plt.ylabel('price')
plt.show()
```



```
sns.barplot(data=df, x='neighbourhood_group', y='price', errorbar=None, es-
timator=np.mean, hue='room_type', palette='Set1')
plt.title("average price by neighbourhood
group", fontsize='15', weight='light')
plt.xlabel("neighbourhood_group")
plt.ylabel('price')
plt.show()
```

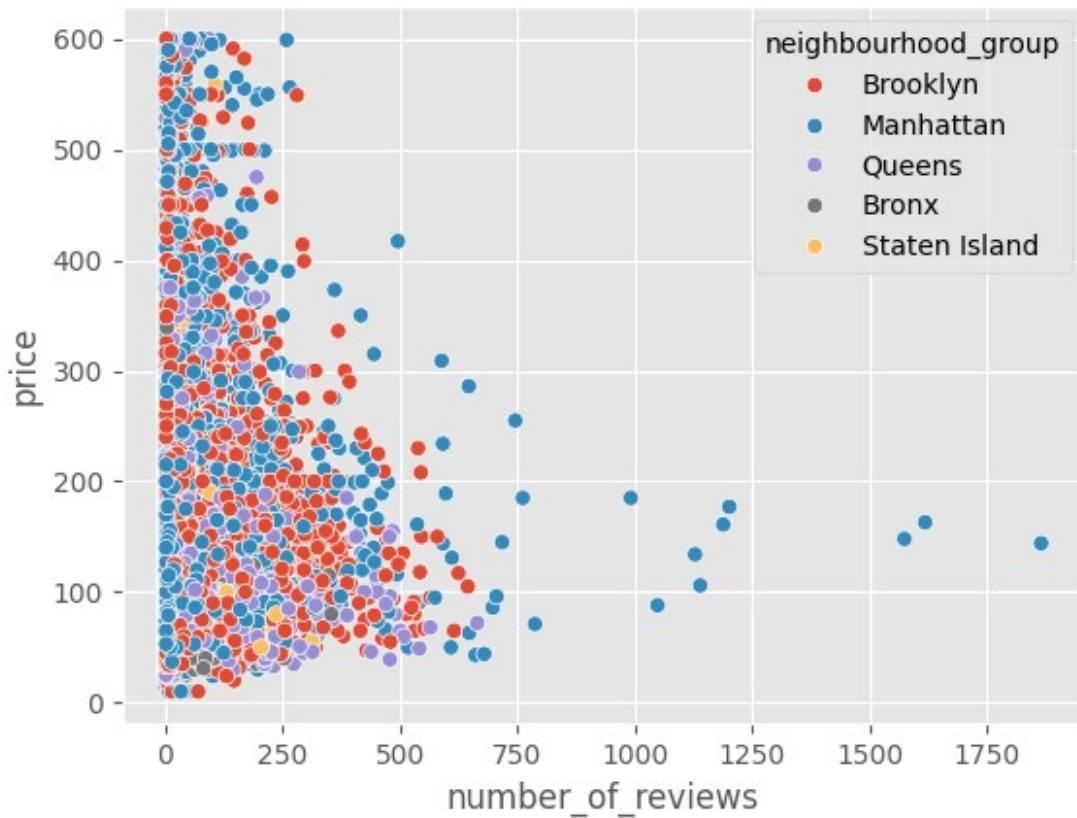


```
sns.barplot(data=df,x='room_type',y='price',errorbar=None,estimator=np
.mean,hue='neighbourhood_group',palette='Set1')
plt.title("average price by neighbourhood
group",fontsize='15',weight='light')
plt.xlabel("neighbourhood_group")
plt.ylabel('price')
plt.show()
```



```
sns.scatterplot(data=df,y='price',x='number_of_reviews',hue='neighbourhood_group')
```

```
<Axes: xlabel='number_of_reviews', ylabel='price'>
```

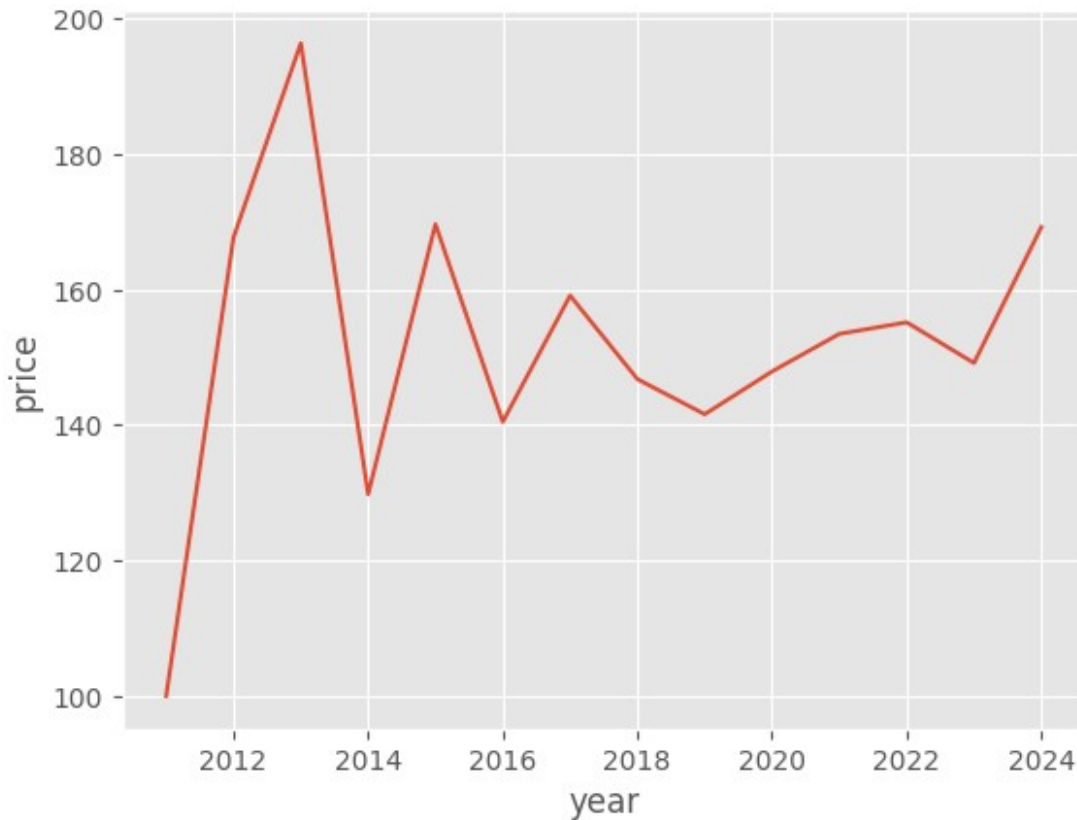


```
df['year']= df['last_review'].dt.year
```

```
#!/ line plot trend over year like avg price per year
```

```
avg_price= df.groupby('year')['price'].mean().reset_index()  
sns.lineplot(data=avg_price,x='year',y='price',palette='Set3')
```

```
<Axes: xlabel='year', ylabel='price'>
```

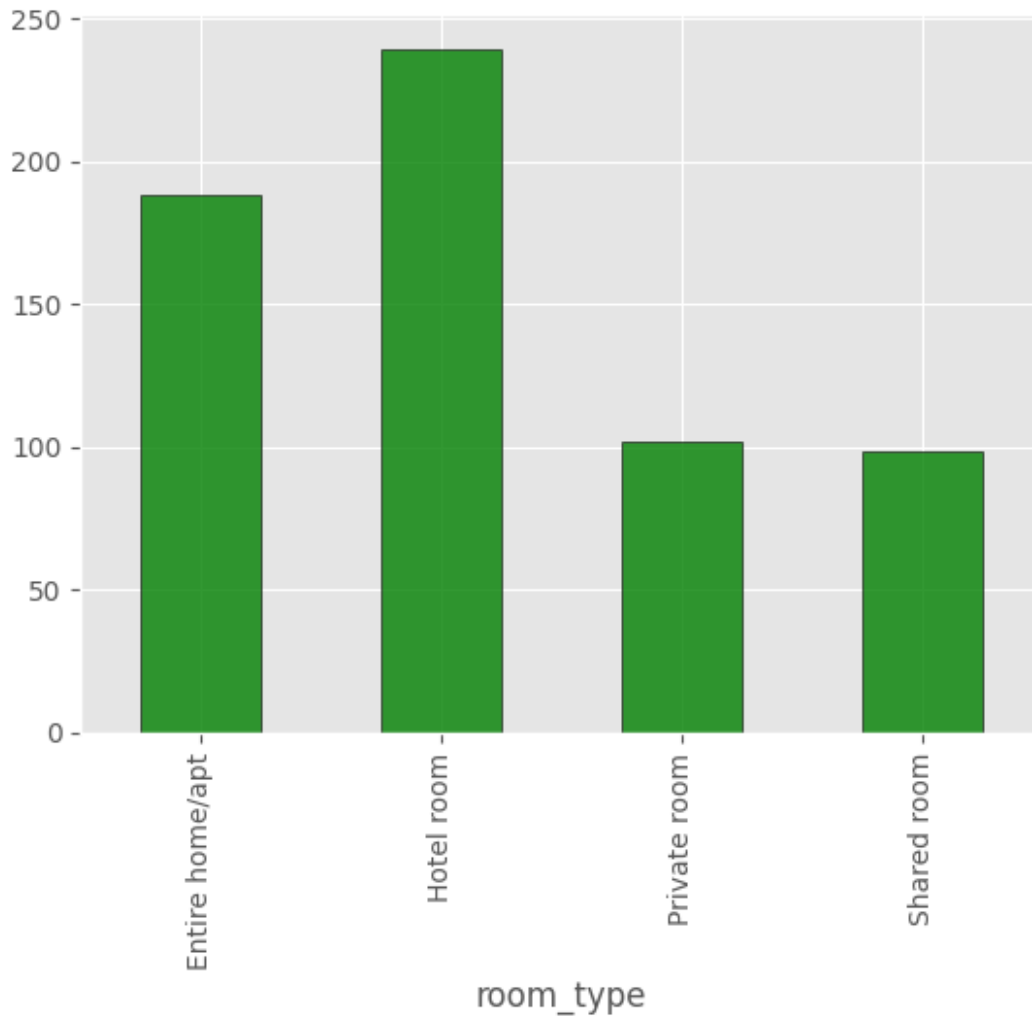


```
df.groupby('room_type')['price'].mean()
```

```
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    [
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    ],
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      "98.58245614035087"
    ]
  ],
  "shape": {
    "columns": 1,
    "rows": 4
  }
}
```

```
df.groupby('room_type')
['price'].mean().plot(kind='bar', color='green', alpha=0.8, edgecolor='black')
```

```
<Axes: xlabel='room_type'>
```



```
# sns.pairplot(data=df, hue='neighbourhood_group')
```

Multivariate analysis

df

```
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```

```

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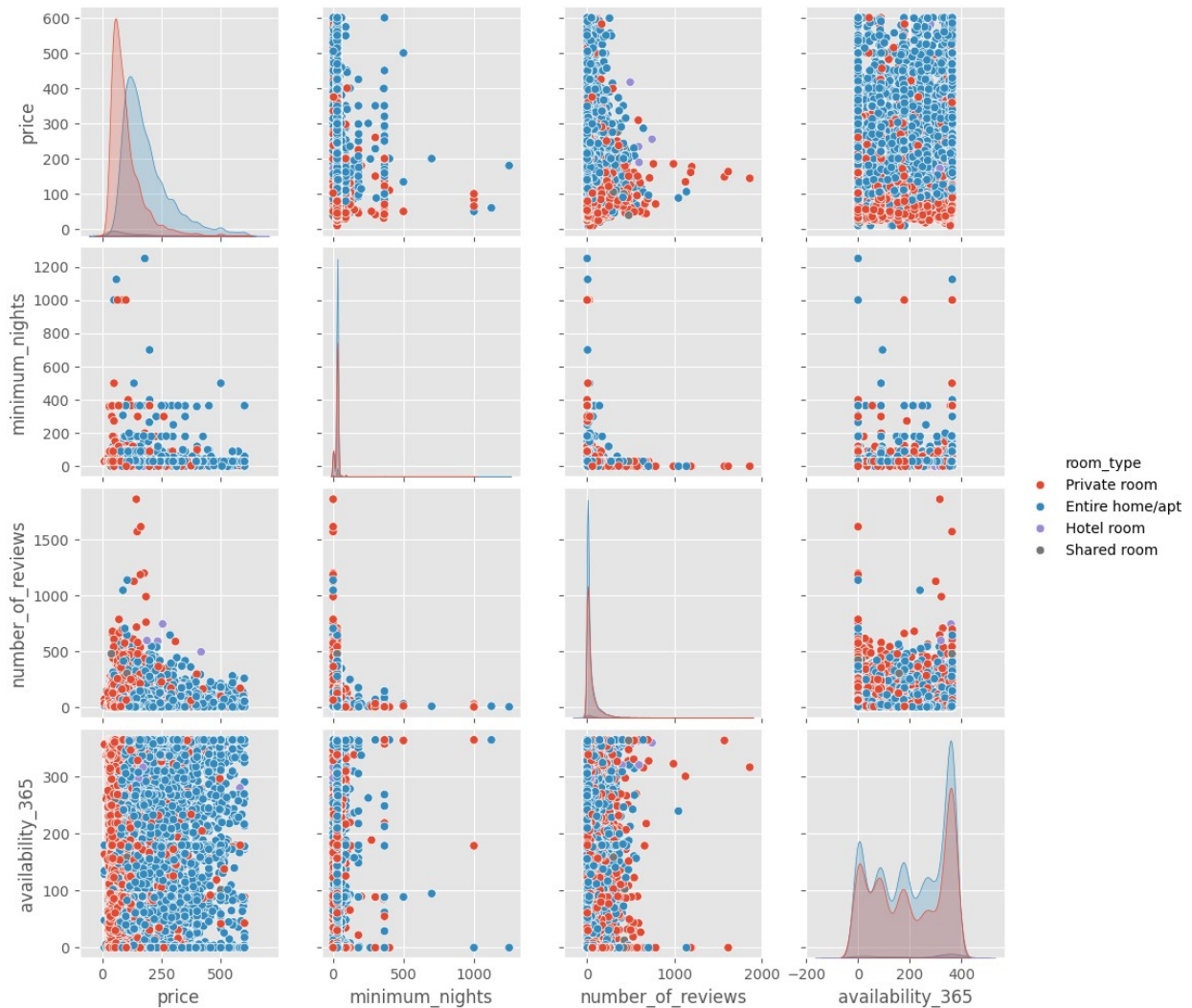
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home/apt","75","30.0","35.0","2023-10-10
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<Figure size 1000x1000 with 0 Axes>



<Figure size 1000x1000 with 0 Axes>

```
df.columns
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```
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       'reviews_per_month', 'calculated_host_listings_count',
       'availability_365', 'number_of_reviews_ltm', 'license',
       'rating',
       'bedrooms', 'beds', 'baths', 'per_bed', 'year'],
      dtype='object')
```

```
## geographical distrubutation of airbnb listing
```

```
plt.figure(figsize=(10,10))
```

```
sns.scatterplot(data=df,x='longitude',y='latitude',hue='room_type')
```

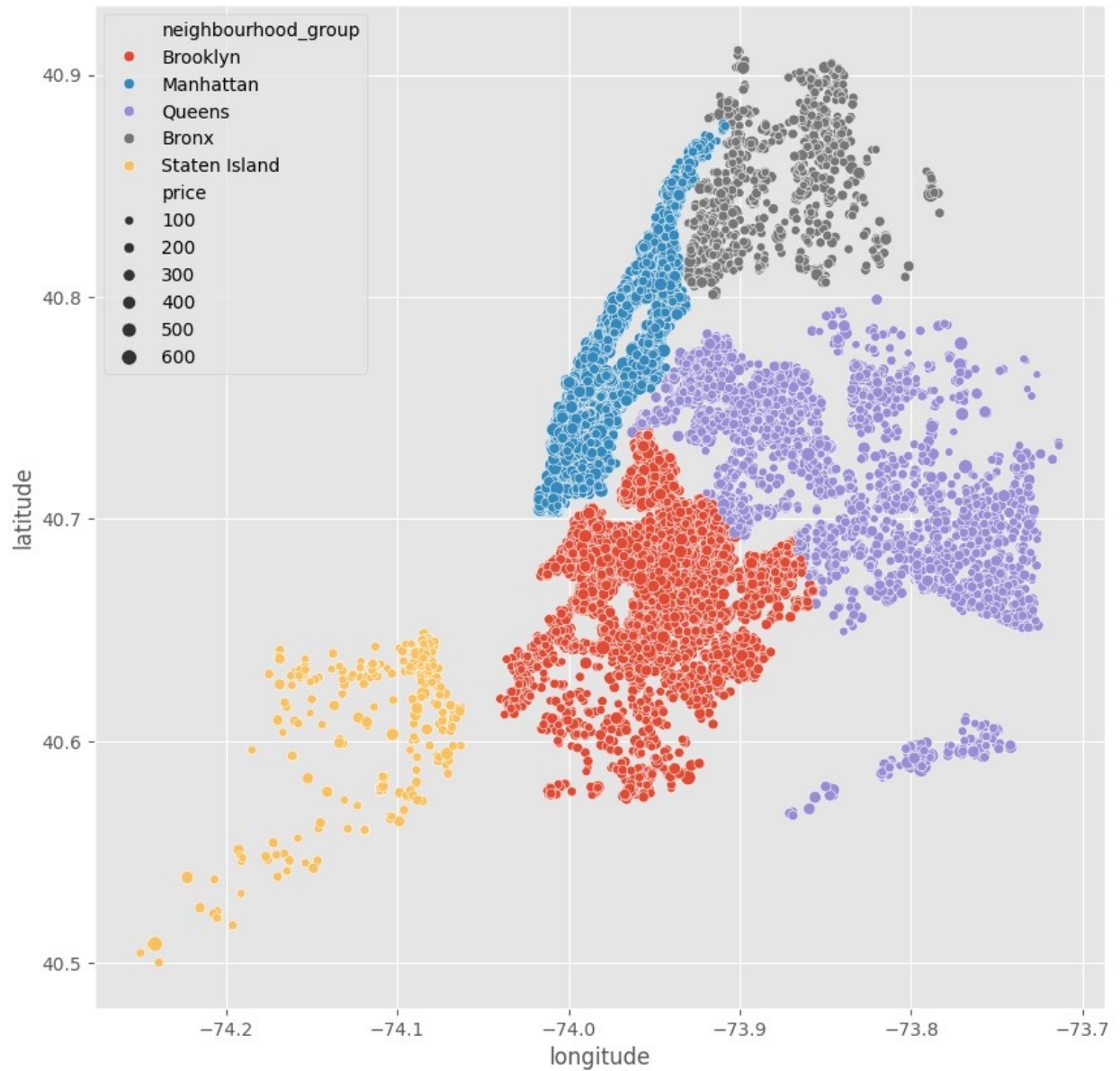


```
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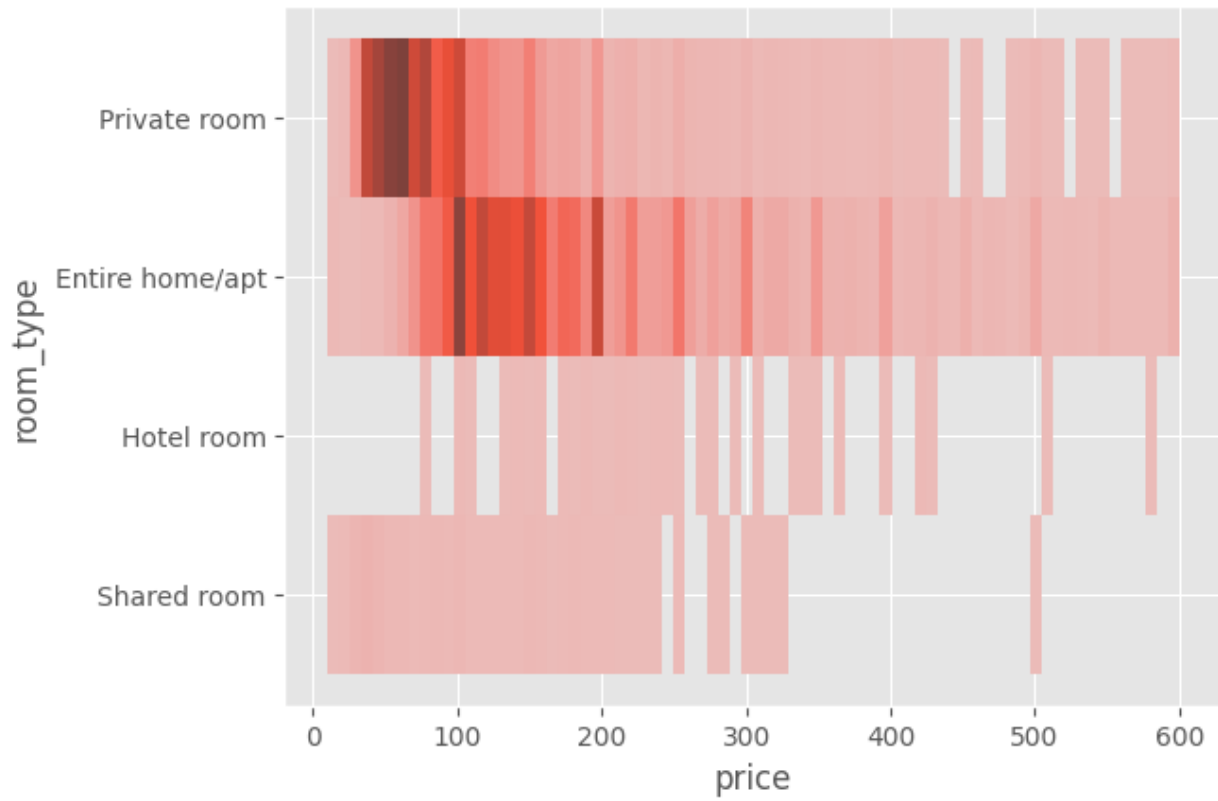


```
plt.figure(figsize=(10,10))
sns.scatterplot(data=df,x='longitude',y='latitude',hue='neighbourhood_group',size='price')
```

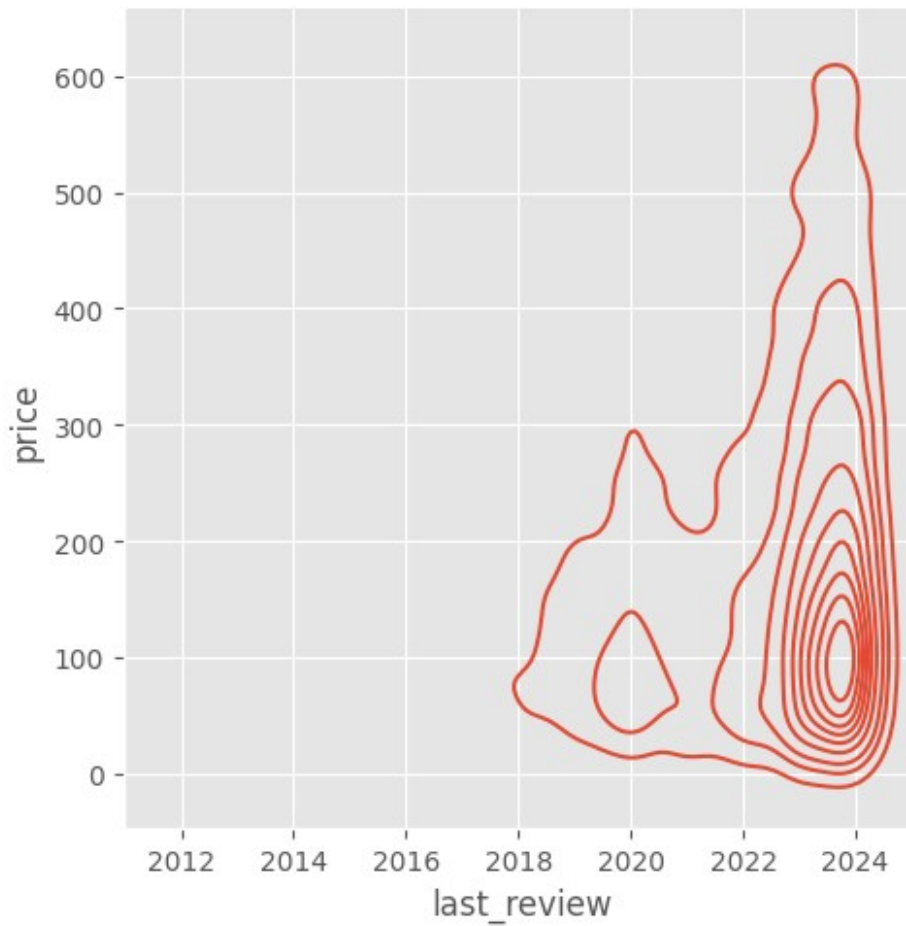
```
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```

```
sns.histplot(data=df,x='price',y='room_type')  
<Axes: xlabel='price', ylabel='room_type'>
```

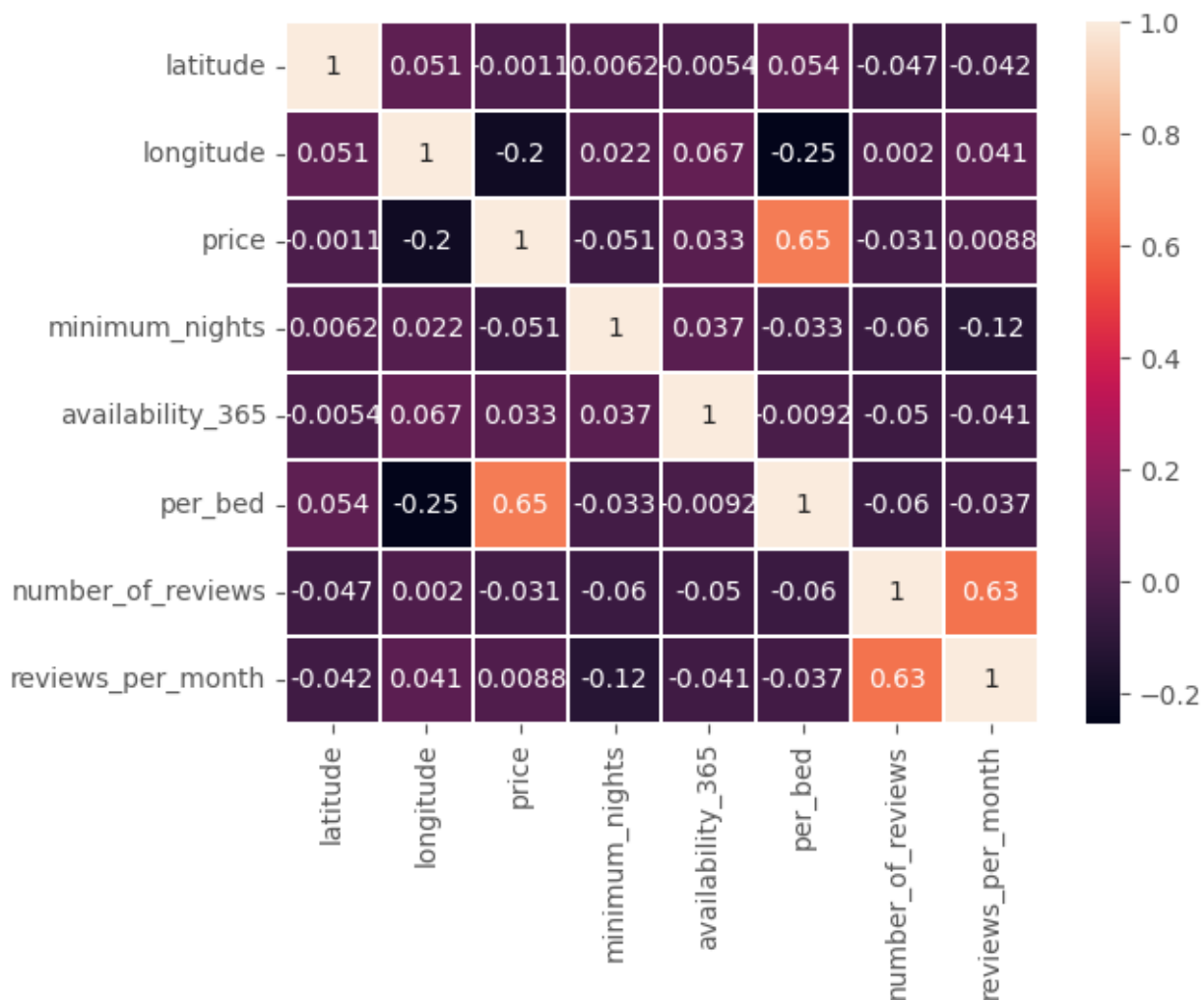


```
sns.displot(data=df,x='last_review',y='price',kind='kde')  
<seaborn.axisgrid.FacetGrid at 0x2883faeeba0>
```



heatmap

```
corr=df[['latitude','longitude','price','minimum_nights','availability_365','per_bed','number_of_reviews','reviews_per_month']].corr()
sns.heatmap(data=corr,annot=True,linewidths=0.1)
<Axes: >
```



df

```
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```

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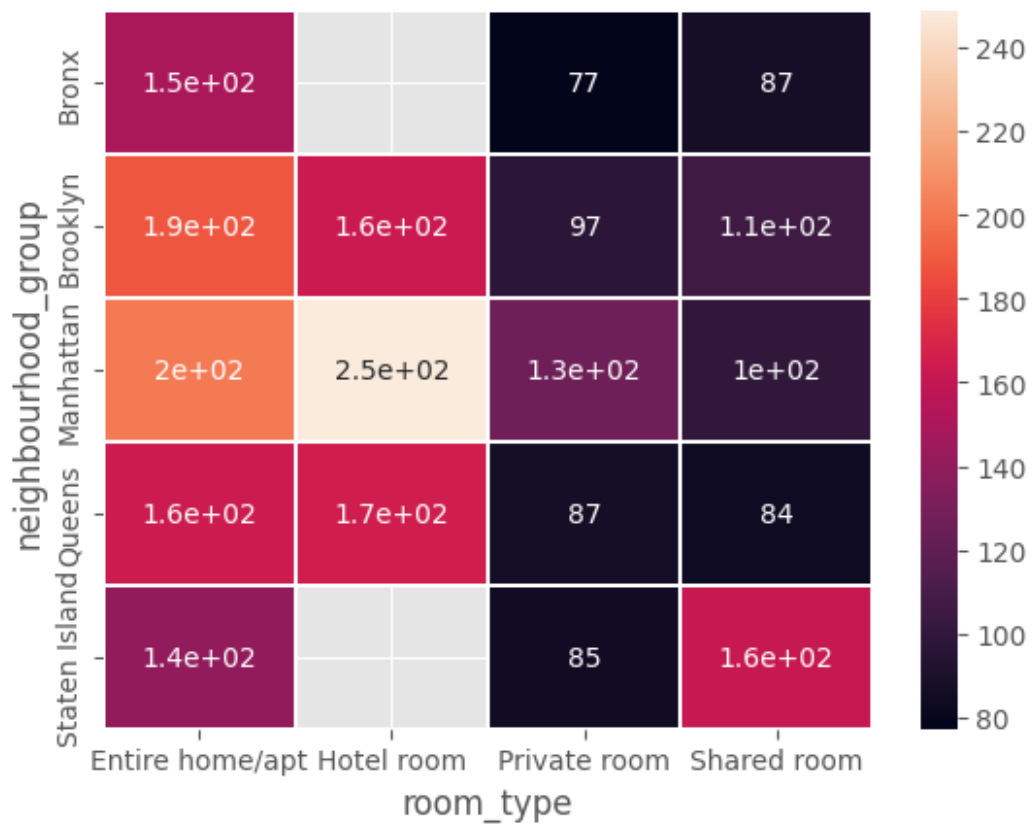
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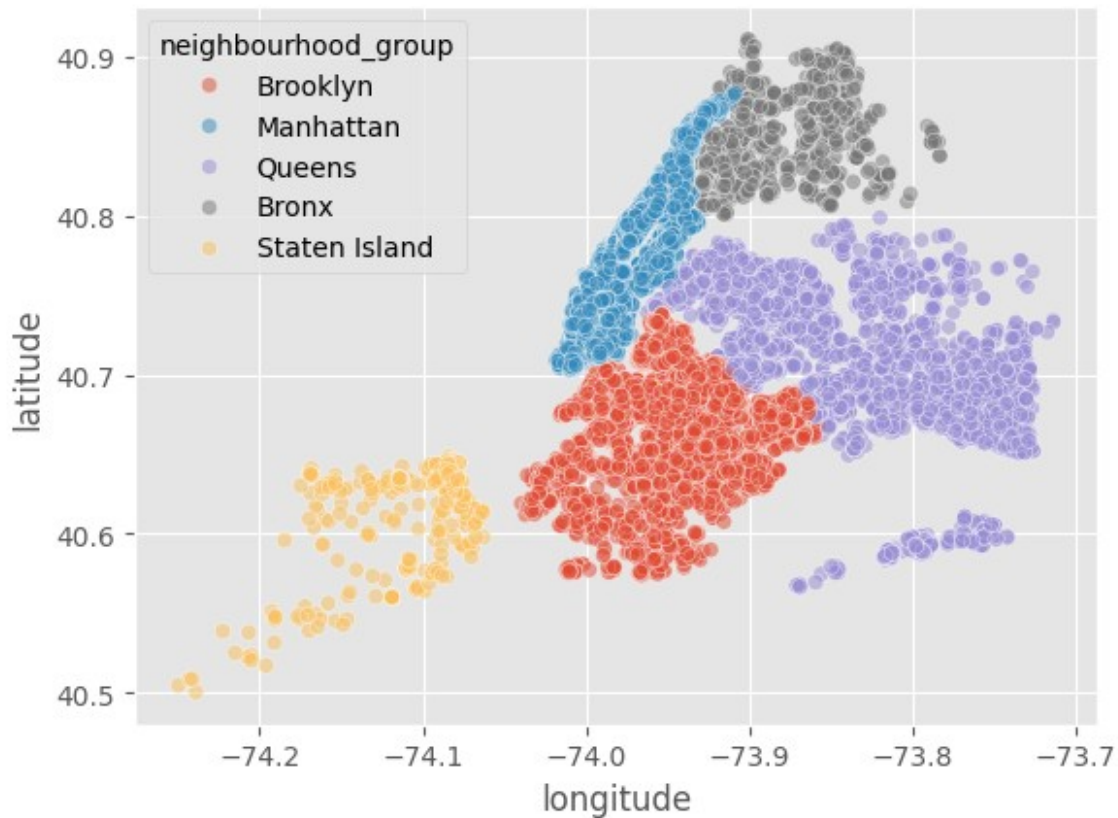
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df

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